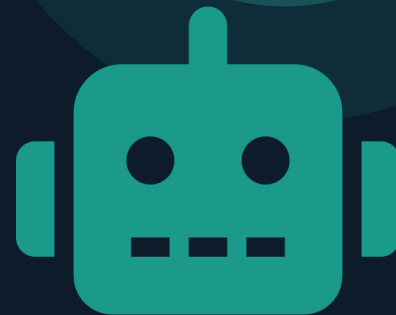


CHAPTER 1

Introduction to Robotics

Sensors · Controllers · Actuators · Control Systems



What is a Robot?

Definition

A robot is an autonomous or semi-autonomous machine that performs tasks automatically by:



Sensing

Cameras, touch sensors, ultrasonic



Processing

Computer programs / controllers



Acting

Motors, arms, wheels

Key Characteristics



Autonomous or semi-autonomous operation



Can sense the environment

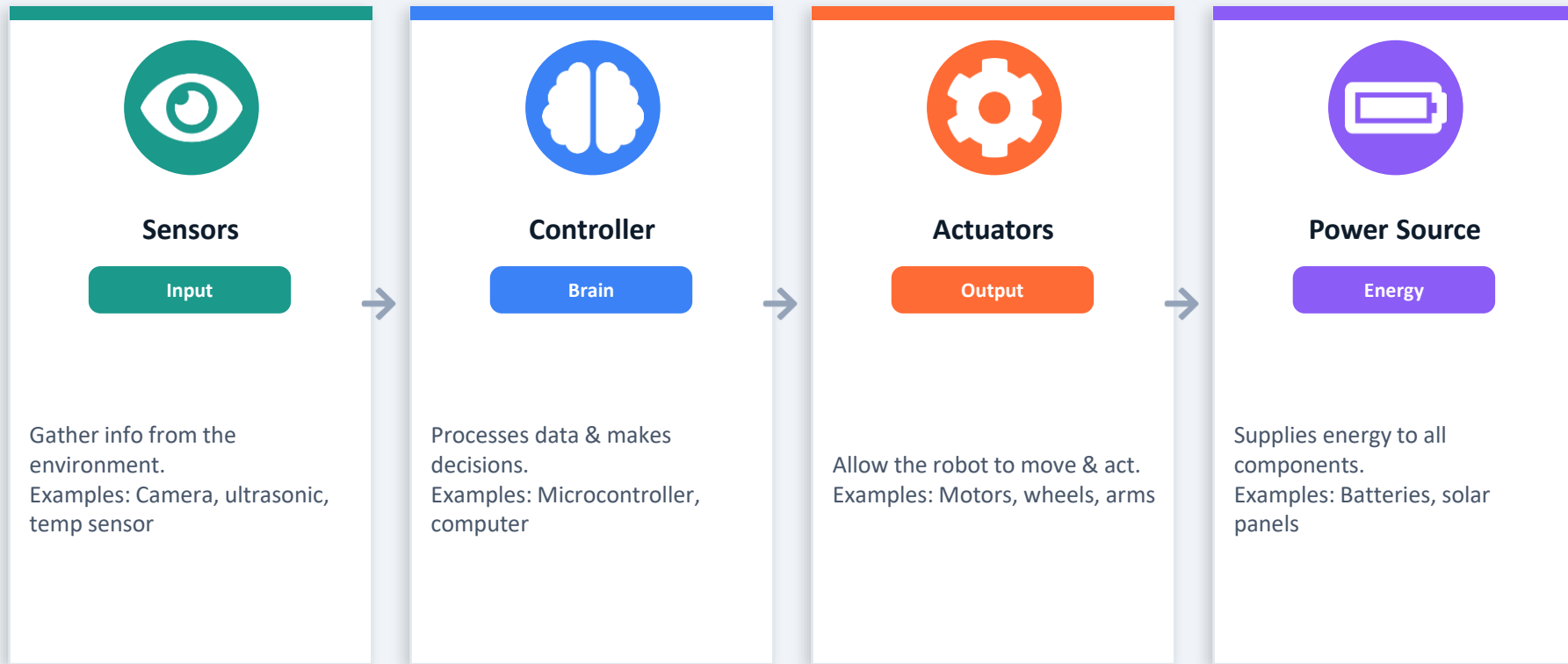


Can process information



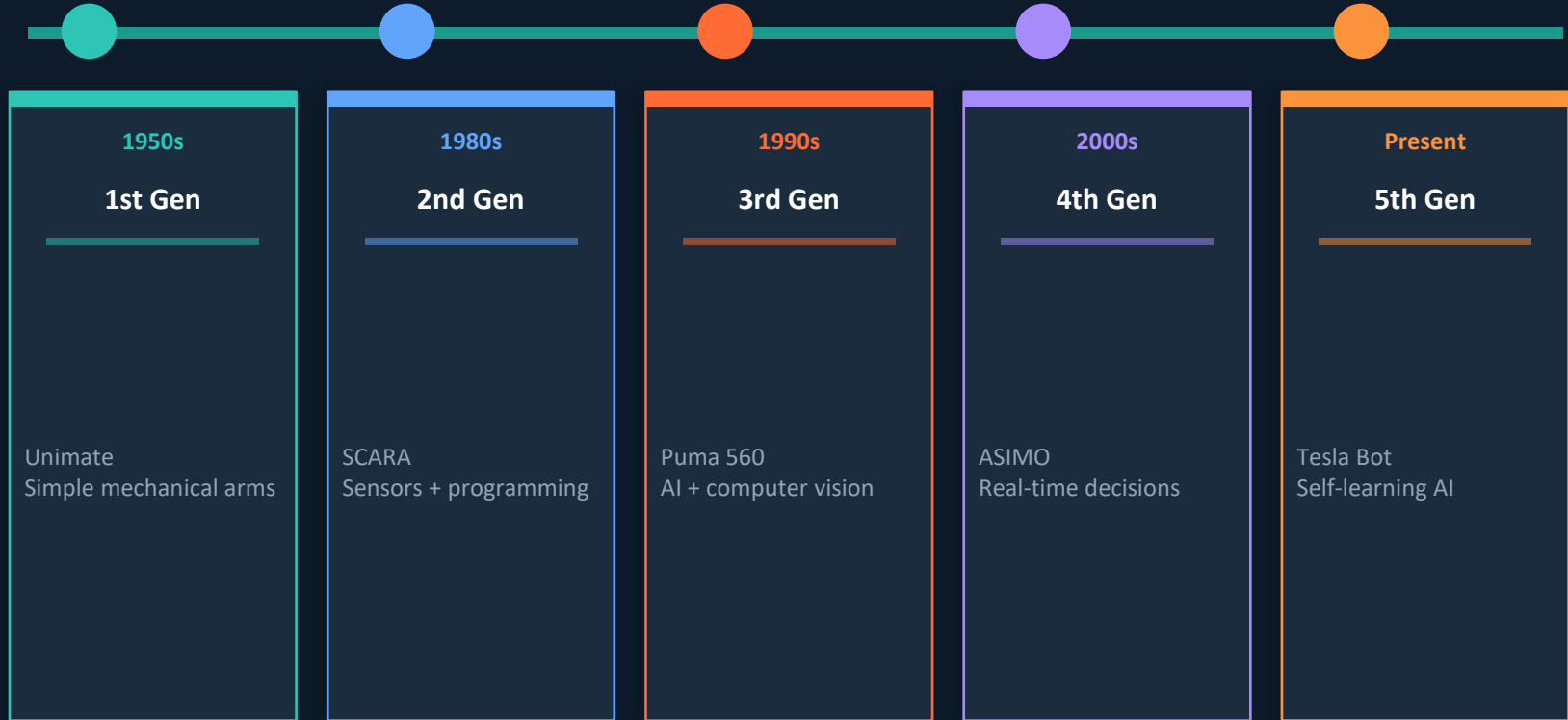
Can take actions based on decisions

Components of a Robotic System



History & Evolution of Robotics

Five generations of progress



The IPO Model — How Robots Work

INPUT



Sensors collect data from the environment

Example:

Roomba's dirt sensor detects dirt on floor

PROCESS



Controller analyzes data and decides

Example:

Controller maps the best cleaning path

OUTPUT



Actuators carry out physical actions

Example:

Wheels move to clean the area

Control Systems

OPEN-LOOP (No Feedback)

Actions are pre-set and run without checking if the goal was achieved.



Example: Microwave Oven

No feedback — it does NOT check if food is actually hot.

Weakness

Cannot adapt to errors or changes

VS

CLOSED-LOOP (With Feedback)

Continuously checks its output and adjusts in real time.



Example: Air Conditioner with Thermostat

Constantly checks room temp and adjusts to hit 22°C.

Strength

Self-correcting and adaptive

Applications of Robotics

Where robots are transforming the world

Industry



Automates dangerous & repetitive tasks — car assembly, welding, painting

Healthcare



Surgical robots for precision ops; assistive robots for elderly & rehabilitation

Exploration



Operates in extreme environments — space rovers, deep-sea research robots

Everyday Life



Simplifies chores (Roomba), education robots, AI assistants, delivery drones

⚠ Ethical Debates: Job displacement · Moral decision-making · Military use risks

Lesson Summary

Chapter 1: Introduction to Robotics



A robot senses, processes, and acts — all three are essential to its function.



The four components are Sensors, Controller, Actuators, and Power Source.



Robotics has evolved through 5 generations — from simple arms to self-learning AI.



The IPO model (Input → Process → Output) describes every robot's operation.



Closed-loop systems self-correct using feedback; open-loop systems cannot adapt.